



RANDOM VARIABLES AND EXPECTATION

4.1 RANDOM VARIABLES

When a random experiment is performed, we are often not interested in all of the details of the experimental result but only in the value of some numerical quantity determined by the result. For instance, in tossing dice we are often interested in the sum of the two dice and are not really concerned about the values of the individual dice. That is, we may be interested in knowing that the sum is 7 and not be concerned over whether the actual outcome was (1, 6) or (2, 5) or (3, 4) or (4, 3) or (5, 2) or (6, 1). Also, a civil engineer may not be directly concerned with the daily risings and declines of the water level of a reservoir (which we can take as the experimental result) but may only care about the level at the end of a rainy season. These quantities of interest that are determined by the result of the experiment are known as *random variables*.

Since the value of a random variable is determined by the outcome of the experiment, we may assign probabilities of its possible values.

EXAMPLE 4.1a Letting X denote the random variable that is defined as the sum of two fair dice, then

$$P\{X = 2\} = P\{(1, 1)\} = \frac{1}{36} \quad (4.1.1)$$

$$P\{X = 3\} = P\{(1, 2), (2, 1)\} = \frac{2}{36}$$

$$P\{X = 4\} = P\{(1, 3), (2, 2), (3, 1)\} = \frac{3}{36}$$

$$P\{X = 5\} = P\{(1, 4), (2, 3), (3, 2), (4, 1)\} = \frac{4}{36}$$

$$P\{X = 6\} = P\{(1, 5), (2, 4), (3, 3), (4, 2), (5, 1)\} = \frac{5}{36}$$

$$P\{X = 7\} = P\{(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)\} = \frac{6}{36}$$

$$P\{X = 8\} = P\{(2, 6), (3, 5), (4, 4), (5, 3), (6, 2)\} = \frac{5}{36}$$

$$\begin{aligned}
P\{X = 9\} &= P\{(3, 6), (4, 5), (5, 4), (6, 3)\} = \frac{4}{36} \\
P\{X = 10\} &= P\{(4, 6), (5, 5), (6, 4)\} = \frac{3}{36} \\
P\{X = 11\} &= P\{(5, 6), (6, 5)\} = \frac{2}{36} \\
P\{X = 12\} &= P\{(6, 6)\} = \frac{1}{36}
\end{aligned}$$

In other words, the random variable X can take on any integral value between 2 and 12 and the probability that it takes on each value is given by Equation 4.1.1. Since X must take on some value, we must have

$$1 = P(S) = P\left(\bigcup_{i=2}^{12} \{X = i\}\right) = \sum_{i=2}^{12} P\{X = i\}$$

which is easily verified from Equation 4.1.1.

Another random variable of possible interest in this experiment is the value of the first die. Letting Y denote this random variable, then Y is equally likely to take on any of the values 1 through 6. That is,

$$P\{Y = i\} = 1/6, \quad i = 1, 2, 3, 4, 5, 6 \quad \blacksquare$$

EXAMPLE 4.1b Suppose that an individual purchases two electronic components, each of which may be either defective or acceptable. In addition, suppose that the four possible results — (d, d) , (d, a) , (a, d) , (a, a) — have respective probabilities .09, .21, .21, .49 [where (d, d) means that both components are defective, (d, a) that the first component is defective and the second acceptable, and so on]. If we let X denote the number of acceptable components obtained in the purchase, then X is a random variable taking on one of the values 0, 1, 2 with respective probabilities

$$\begin{aligned}
P\{X = 0\} &= .09 \\
P\{X = 1\} &= .42 \\
P\{X = 2\} &= .49
\end{aligned}$$

If we were mainly concerned with whether there was at least one acceptable component, we could define the random variable I by

$$I = \begin{cases} 1 & \text{if } X = 1 \text{ or } 2 \\ 0 & \text{if } X = 0 \end{cases}$$

If A denotes the event that at least one acceptable component is obtained, then the random variable I is called the *indicator* random variable for the event A , since I will equal 1

or 0 depending upon whether A occurs. The probabilities attached to the possible values of I are

$$P\{I=1\} = .91$$

$$P\{I=0\} = .09 \quad \blacksquare$$

In the two foregoing examples, the random variables of interest took on a finite number of possible values. Random variables whose set of possible values can be written either as a finite sequence $x_1, \dots, x_n$, or as an infinite sequence $x_1, \dots$ are said to be *discrete*. For instance, a random variable whose set of possible values is the set of nonnegative integers is a discrete random variable. However, there also exist random variables that take on a continuum of possible values. These are known as *continuous* random variables. One example is the random variable denoting the lifetime of a car, when the car's lifetime is assumed to take on any value in some interval (a, b) .

The *cumulative distribution function*, or more simply the *distribution function*, F of the random variable X is defined for any real number x by

$$F(x) = P\{X \leq x\}$$

That is, $F(x)$ is the probability that the random variable X takes on a value that is less than or equal to x .

Notation: We will use the notation $X \sim F$ to signify that F is the distribution function of X .

All probability questions about X can be answered in terms of its distribution function F . For example, suppose we wanted to compute $P\{a < X \leq b\}$. This can be accomplished by first noting that the event $\{X \leq b\}$ can be expressed as the union of the two mutually exclusive events $\{X \leq a\}$ and $\{a < X \leq b\}$. Therefore, applying Axiom 3, we obtain that

$$P\{X \leq b\} = P\{X \leq a\} + P\{a < X \leq b\}$$

or

$$P\{a < X \leq b\} = F(b) - F(a)$$

EXAMPLE 4.1c Suppose the random variable X has distribution function

$$F(x) = \begin{cases} 0 & x \leq 0 \\ 1 - \exp\{-x^2\} & x > 0 \end{cases}$$

What is the probability that X exceeds 1?

SOLUTION The desired probability is computed as follows:

$$\begin{aligned}
 P\{X > 1\} &= 1 - P\{X \leq 1\} \\
 &= 1 - F(1) \\
 &= e^{-1} \\
 &= .368 \quad \blacksquare
 \end{aligned}$$

4.2 TYPES OF RANDOM VARIABLES

As was previously mentioned, a random variable whose set of possible values is a sequence is said to be *discrete*. For a discrete random variable X , we define the *probability mass function* $p(a)$ of X by

$$p(a) = P\{X = a\}$$

The probability mass function $p(a)$ is positive for at most a countable number of values of a . That is, if X must assume one of the values $x_1, x_2, \dots$, then

$$\begin{aligned}
 p(x_i) &> 0, & i = 1, 2, \dots \\
 p(x) &= 0, & \text{all other values of } x
 \end{aligned}$$

Since X must take on one of the values x_i , we have

$$\sum_{i=1}^{\infty} p(x_i) = 1$$

EXAMPLE 4.2a Consider a random variable X that is equal to 1, 2, or 3. If we know that

$$p(1) = \frac{1}{2} \quad \text{and} \quad p(2) = \frac{1}{3}$$

then it follows (since $p(1) + p(2) + p(3) = 1$) that

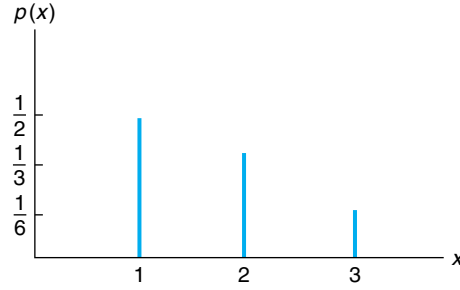
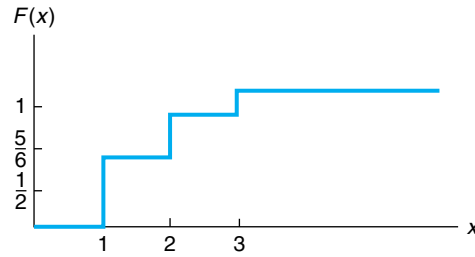
$$p(3) = \frac{1}{6}$$

A graph of $p(x)$ is presented in Figure 4.1. ■

The cumulative distribution function F can be expressed in terms of $p(x)$ by

$$F(a) = \sum_{\text{all } x \leq a} p(x)$$

If X is a discrete random variable whose set of possible values are $x_1, x_2, x_3, \dots$, where $x_1 < x_2 < x_3 < \dots$, then its distribution function F is a step function. That is, the value of F is constant in the intervals $[x_{i-1}, x_i)$ and then takes a step (or jump) of size $p(x_i)$ at x_i .

FIGURE 4.1 Graph of $p(x)$, Example 4.2a.FIGURE 4.2 Graph of $F(x)$.

For instance, suppose X has a probability mass function given (as in Example 4.2a) by

$$p(1) = \frac{1}{2}, \quad p(2) = \frac{1}{3}, \quad p(3) = \frac{1}{6}$$

Then the cumulative distribution function F of X is given by

$$F(a) = \begin{cases} 0 & a < 1 \\ \frac{1}{2} & 1 \leq a < 2 \\ \frac{5}{6} & 2 \leq a < 3 \\ 1 & 3 \leq a \end{cases}$$

This is graphically presented in Figure 4.2.

Whereas the set of possible values of a discrete random variable is a sequence, we often must consider random variables whose set of possible values is an interval. Let X be such a random variable. We say that X is a *continuous* random variable if there exists a nonnegative function $f(x)$, defined for all real $x \in (-\infty, \infty)$, having the property that for any set B of real numbers

$$P\{X \in B\} = \int_B f(x) dx \quad (4.2.1)$$

The function $f(x)$ is called the *probability density function* of the random variable X .

In words, Equation 4.2.1 states that the probability that X will be in B may be obtained by integrating the probability density function over the set B . Since X must assume some value, $f(x)$ must satisfy

$$1 = P\{X \in (-\infty, \infty)\} = \int_{-\infty}^{\infty} f(x) dx$$

All probability statements about X can be answered in terms of $f(x)$. For instance, letting $B = [a, b]$, we obtain from Equation 4.2.1 that

$$P\{a \leq X \leq b\} = \int_a^b f(x) dx \quad (4.2.2)$$

If we let $a = b$ in the above, then

$$P\{X = a\} = \int_a^a f(x) dx = 0$$

In words, this equation states that the probability that a continuous random variable will assume any *particular* value is zero. (See Figure 4.3.)

The relationship between the cumulative distribution $F(\cdot)$ and the probability density $f(\cdot)$ is expressed by

$$F(a) = P\{X \in (-\infty, a]\} = \int_{-\infty}^a f(x) dx$$

Differentiating both sides yields

$$\frac{d}{da} F(a) = f(a)$$

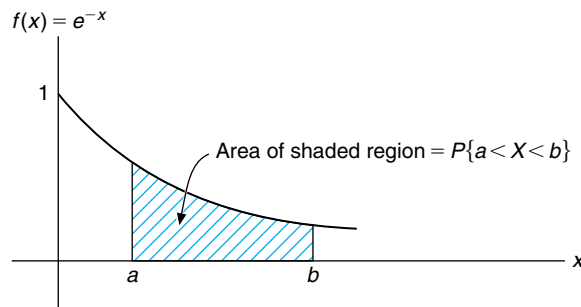


FIGURE 4.3 The probability density function $f(x) = \begin{cases} e^{-x} & x \geq 0 \\ 0 & x < 0 \end{cases}$.

That is, the density is the derivative of the cumulative distribution function. A somewhat more intuitive interpretation of the density function may be obtained from Equation 4.2.2 as follows:

$$P\left\{a - \frac{\varepsilon}{2} \leq X \leq a + \frac{\varepsilon}{2}\right\} = \int_{a-\varepsilon/2}^{a+\varepsilon/2} f(x) dx \approx \varepsilon f(a)$$

when ε is small. In other words, the probability that X will be contained in an interval of length ε around the point a is approximately $\varepsilon f(a)$. From this, we see that $f(a)$ is a measure of how likely it is that the random variable will be near a .

EXAMPLE 4.2b Suppose that X is a continuous random variable whose probability density function is given by

$$f(x) = \begin{cases} C(4x - 2x^2) & 0 < x < 2 \\ 0 & \text{otherwise} \end{cases}$$

- (a) What is the value of C ?
- (b) Find $P\{X > 1\}$.

SOLUTION (a) Since f is a probability density function, we must have that $\int_{-\infty}^{\infty} f(x) dx = 1$, implying that

$$C \int_0^2 (4x - 2x^2) dx = 1$$

or

$$C \left[2x^2 - \frac{2x^3}{3} \right] \Big|_{x=0}^{x=2} = 1$$

or

$$C = \frac{3}{8}$$

(b) Hence

$$P\{X > 1\} = \int_1^{\infty} f(x) dx = \frac{3}{8} \int_1^2 (4x - 2x^2) dx = \frac{1}{2} \quad \blacksquare$$

4.3 JOINTLY DISTRIBUTED RANDOM VARIABLES

For a given experiment, we are often interested not only in probability distribution functions of individual random variables but also in the relationships between two or more random variables. For instance, in an experiment into the possible causes of cancer, we might

be interested in the relationship between the average number of cigarettes smoked daily and the age at which an individual contracts cancer. Similarly, an engineer might be interested in the relationship between the shear strength and the diameter of a spot weld in a fabricated sheet steel specimen.

To specify the relationship between two random variables, we define the joint cumulative probability distribution function of X and Y by

$$F(x, y) = P\{X \leq x, Y \leq y\}$$

A knowledge of the joint probability distribution function enables one, at least in theory, to compute the probability of any statement concerning the values of X and Y . For instance, the distribution function of X — call it F_X — can be obtained from the joint distribution function F of X and Y as follows:

$$\begin{aligned} F_X(x) &= P\{X \leq x\} \\ &= P\{X \leq x, Y < \infty\} \\ &= F(x, \infty) \end{aligned}$$

Similarly, the cumulative distribution function of Y is given by

$$F_Y(y) = F(\infty, y)$$

In the case where X and Y are both discrete random variables whose possible values are, respectively, $x_1, x_2, \dots$, and $y_1, y_2, \dots$, we define the *joint probability mass function* of X and Y , $p(x_i, y_j)$, by

$$p(x_i, y_j) = P\{X = x_i, Y = y_j\}$$

The individual probability mass functions of X and Y are easily obtained from the joint probability mass function by the following reasoning. Since Y must take on some value y_j , it follows that the event $\{X = x_i\}$ can be written as the union, over all j , of the mutually exclusive events $\{X = x_i, Y = y_j\}$. That is,

$$\{X = x_i\} = \bigcup_j \{X = x_i, Y = y_j\}$$

and so, using Axiom 3 of the probability function, we see that

$$\begin{aligned} P\{X = x_i\} &= P\left(\bigcup_j \{X = x_i, Y = y_j\}\right) \\ &= \sum_j P\{X = x_i, Y = y_j\} \\ &= \sum_j p(x_i, y_j) \end{aligned} \tag{4.3.1}$$

Similarly, we can obtain $P\{Y = y_j\}$ by summing $p(x_i, y_j)$ over all possible values of x_i , that is,

$$\begin{aligned} P\{Y = y_j\} &= \sum_i P\{X = x_i, Y = y_j\} \\ &= \sum_i p(x_i, y_j) \end{aligned} \quad (4.3.2)$$

Hence, specifying the joint probability mass function always determines the individual mass functions. However, it should be noted that the reverse is not true. Namely, knowledge of $P\{X = x_i\}$ and $P\{Y = y_j\}$ does not determine the value of $P\{X = x_i, Y = y_j\}$.

EXAMPLE 4.3a Suppose that 3 batteries are randomly chosen from a group of 3 new, 4 used but still working, and 5 defective batteries. If we let X and Y denote, respectively, the number of new and used but still working batteries that are chosen, then the joint probability mass function of X and Y , $p(i, j) = P\{X = i, Y = j\}$, is given by

$$\begin{aligned} p(0, 0) &= \binom{5}{3} / \binom{12}{3} = 10/220 \\ p(0, 1) &= \binom{4}{1} \binom{5}{2} / \binom{12}{3} = 40/220 \\ p(0, 2) &= \binom{4}{2} \binom{5}{1} / \binom{12}{3} = 30/220 \\ p(0, 3) &= \binom{4}{3} / \binom{12}{3} = 4/220 \\ p(1, 0) &= \binom{3}{1} \binom{5}{2} / \binom{12}{3} = 30/220 \\ p(1, 1) &= \binom{3}{1} \binom{4}{1} \binom{5}{1} / \binom{12}{3} = 60/220 \\ p(1, 2) &= \binom{3}{1} \binom{4}{2} / \binom{12}{3} = 18/220 \\ p(2, 0) &= \binom{3}{2} \binom{5}{1} / \binom{12}{3} = 15/220 \\ p(2, 1) &= \binom{3}{2} \binom{4}{1} / \binom{12}{3} = 12/220 \\ p(3, 0) &= \binom{3}{3} / \binom{12}{3} = 1/220 \end{aligned}$$

These probabilities can most easily be expressed in tabular form as shown in Table 4.1.

TABLE 4.1 $P\{X = i, Y = j\}$

$i \backslash j$	0	1	2	3	Row Sum $= P\{X = i\}$
0	$\frac{10}{220}$	$\frac{40}{220}$	$\frac{30}{220}$	$\frac{4}{220}$	$\frac{84}{220}$
1	$\frac{30}{220}$	$\frac{60}{220}$	$\frac{18}{220}$	0	$\frac{108}{220}$
2	$\frac{15}{220}$	$\frac{12}{220}$	0	0	$\frac{27}{220}$
3	$\frac{1}{220}$	0	0	0	$\frac{1}{220}$
Column Sums = $P\{Y = j\}$	$\frac{56}{220}$	$\frac{112}{220}$	$\frac{48}{220}$	$\frac{4}{220}$	

TABLE 4.2 $P\{B = i, G = j\}$

$i \backslash j$	0	1	2	3	Row Sum $= P\{B = i\}$
0	.15	.10	.0875	.0375	.3750
1	.10	.175	.1125	0	.3875
2	.0875	.1125	0	0	.2000
3	.0375	0	0	0	.0375
Column Sum = $P\{G = j\}$	.3750	.3875	.2000	.0375	

The reader should note that the probability mass function of X is obtained by computing the row sums, in accordance with the Equation 4.3.1, whereas the probability mass function of Y is obtained by computing the column sums, in accordance with Equation 4.3.2. Because the individual probability mass functions of X and Y thus appear in the margin of such a table, they are often referred to as being the marginal probability mass functions of X and Y , respectively. It should be noted that to check the correctness of such a table we could sum the marginal row (or the marginal column) and verify that its sum is 1. (Why must the sum of the entries in the marginal row (or column) equal 1?) ■

EXAMPLE 4.3b Suppose that 15 percent of the families in a certain community have no children, 20 percent have 1, 35 percent have 2, and 30 percent have 3 children; suppose further that each child is equally likely (and independently) to be a boy or a girl. If a family is chosen at random from this community, then B , the number of boys, and G , the number of girls, in this family will have the joint probability mass function shown in Table 4.2.

These probabilities are obtained as follows:

$$\begin{aligned}
 P\{B = 0, G = 0\} &= P\{\text{no children}\} \\
 &= .15 \\
 P\{B = 0, G = 1\} &= P\{1 \text{ girl and total of 1 child}\} \\
 &= P\{1 \text{ child}\}P\{1 \text{ girl} | 1 \text{ child}\} \\
 &= (.20) \left(\frac{1}{2}\right) = .1 \\
 P\{B = 0, G = 2\} &= P\{2 \text{ girls and total of 2 children}\} \\
 &= P\{2 \text{ children}\}P\{2 \text{ girls} | 2 \text{ children}\} \\
 &= (.35) \left(\frac{1}{2}\right)^2 = .0875 \\
 P\{B = 0, G = 3\} &= P\{3 \text{ girls and total of 3 children}\} \\
 &= P\{3 \text{ children}\}P\{3 \text{ girls} | 3 \text{ children}\} \\
 &= (.30) \left(\frac{1}{2}\right)^3 = .0375
 \end{aligned}$$

We leave it to the reader to verify the remainder of Table 4.2, which tells us, among other things, that the family chosen will have at least 1 girl with probability .625. ■

We say that X and Y are *jointly continuous* if there exists a function $f(x, y)$ defined for all real x and y , having the property that for every set C of pairs of real numbers (that is, C is a set in the two-dimensional plane)

$$P\{(X, Y) \in C\} = \iint_{(x,y) \in C} f(x, y) \, dx \, dy \quad (4.3.3)$$

The function $f(x, y)$ is called the *joint probability density function* of X and Y . If A and B are any sets of real numbers, then by defining $C = \{(x, y) : x \in A, y \in B\}$, we see from Equation 4.3.3 that

$$P\{X \in A, Y \in B\} = \int_B \int_A f(x, y) \, dx \, dy \quad (4.3.4)$$

Because

$$\begin{aligned}
 F(a, b) &= P\{X \in (-\infty, a], Y \in (-\infty, b]\} \\
 &= \int_{-\infty}^b \int_{-\infty}^a f(x, y) \, dx \, dy
 \end{aligned}$$

it follows, upon differentiation, that

$$f(a, b) = \frac{\partial^2}{\partial a \partial b} F(a, b)$$

wherever the partial derivatives are defined. Another interpretation of the joint density function is obtained from Equation 4.3.4 as follows:

$$\begin{aligned} P\{a < X < a + da, b < Y < b + db\} &= \int_b^{b+db} \int_a^{a+da} f(x, y) dx dy \\ &\approx f(a, b) da db \end{aligned}$$

when da and db are small and $f(x, y)$ is continuous at a, b . Hence $f(a, b)$ is a measure of how likely it is that the random vector (X, Y) will be near (a, b) .

If X and Y are jointly continuous, they are individually continuous, and their probability density functions can be obtained as follows:

$$\begin{aligned} P\{X \in A\} &= P\{X \in A, Y \in (-\infty, \infty)\} \\ &= \int_A \int_{-\infty}^{\infty} f(x, y) dy dx \\ &= \int_A f_X(x) dx \end{aligned} \tag{4.3.5}$$

where

$$f_X(x) = \int_{-\infty}^{\infty} f(x, y) dy$$

is thus the probability density function of X . Similarly, the probability density function of Y is given by

$$f_Y(y) = \int_{-\infty}^{\infty} f(x, y) dx \tag{4.3.6}$$

EXAMPLE 4.3c The joint density function of X and Y is given by

$$f(x, y) = \begin{cases} 2e^{-x}e^{-2y} & 0 < x < \infty, 0 < y < \infty \\ 0 & \text{otherwise} \end{cases}$$

Compute (a) $P\{X > 1, Y < 1\}$; (b) $P\{X < Y\}$; and (c) $P\{X < a\}$.

SOLUTION

$$\begin{aligned}
 \text{(a)} \quad P\{X > 1, Y < 1\} &= \int_0^1 \int_1^\infty 2e^{-x} e^{-2y} dx dy \\
 &= \int_0^1 2e^{-2y} (-e^{-x}|_1^\infty) dy \\
 &= e^{-1} \int_0^1 2e^{-2y} dy \\
 &= e^{-1} (1 - e^{-2})
 \end{aligned}$$

$$\begin{aligned}
 \text{(b)} \quad P\{X < Y\} &= \iint_{(x,y): x < y} 2e^{-x} e^{-2y} dx dy \\
 &= \int_0^\infty \int_0^y 2e^{-x} e^{-2y} dx dy \\
 &= \int_0^\infty 2e^{-2y} (1 - e^{-y}) dy \\
 &= \int_0^\infty 2e^{-2y} dy - \int_0^\infty 2e^{-3y} dy \\
 &= 1 - \frac{2}{3} \\
 &= \frac{1}{3}
 \end{aligned}$$

$$\begin{aligned}
 \text{(c)} \quad P\{X < a\} &= \int_0^a \int_0^\infty 2e^{-2y} e^{-x} dy dx \\
 &= \int_0^a e^{-x} dx \\
 &= 1 - e^{-a} \quad \blacksquare
 \end{aligned}$$

4.3.1 INDEPENDENT RANDOM VARIABLES

The random variables X and Y are said to be independent if for any two sets of real numbers A and B

$$P\{X \in A, Y \in B\} = P\{X \in A\}P\{Y \in B\} \quad (4.3.7)$$

In other words, X and Y are independent if, for all A and B , the events $E_A = \{X \in A\}$ and $F_B = \{Y \in B\}$ are independent.

It can be shown by using the three axioms of probability that Equation 4.3.7 will follow if and only if for all a, b

$$P\{X \leq a, Y \leq b\} = P\{X \leq a\}P\{Y \leq b\}$$

Hence, in terms of the joint distribution function F of X and Y , we have that X and Y are independent if

$$F(a, b) = F_X(a)F_Y(b) \quad \text{for all } a, b$$

When X and Y are discrete random variables, the condition of independence Equation 4.3.7 is equivalent to

$$p(x, y) = p_X(x)p_Y(y) \quad \text{for all } x, y \quad (4.3.8)$$

where p_X and p_Y are the probability mass functions of X and Y . The equivalence follows because, if Equation 4.3.7 is satisfied, then we obtain Equation 4.3.8 by letting A and B be, respectively, the one-point sets $A = \{x\}$, $B = \{y\}$. Furthermore, if Equation 4.3.8 is valid, then for any sets A, B

$$\begin{aligned} P\{X \in A, Y \in B\} &= \sum_{y \in B} \sum_{x \in A} p(x, y) \\ &= \sum_{y \in B} \sum_{x \in A} p_X(x)p_Y(y) \\ &= \sum_{y \in B} p_Y(y) \sum_{x \in A} p_X(x) \\ &= P\{Y \in B\}P\{X \in A\} \end{aligned}$$

and thus Equation 4.3.7 is established.

In the jointly continuous case, the condition of independence is equivalent to

$$f(x, y) = f_X(x)f_Y(y) \quad \text{for all } x, y$$

Loosely speaking, X and Y are independent if knowing the value of one does not change the distribution of the other. Random variables that are not independent are said to be dependent.

EXAMPLE 4.3d Suppose that X and Y are independent random variables having the common density function

$$f(x) = \begin{cases} e^{-x} & x > 0 \\ 0 & \text{otherwise} \end{cases}$$

Find the density function of the random variable X/Y .

SOLUTION We start by determining the distribution function of X/Y . For $a > 0$

$$\begin{aligned}
 F_{X/Y}(a) &= P\{X/Y \leq a\} \\
 &= \iint_{x/y \leq a} f(x, y) \, dx \, dy \\
 &= \iint_{x/y \leq a} e^{-x} e^{-y} \, dx \, dy \\
 &= \int_0^\infty \int_0^{ay} e^{-x} e^{-y} \, dx \, dy \\
 &= \int_0^\infty (1 - e^{-ay}) e^{-y} \, dy \\
 &= \left[-e^{-y} + \frac{e^{-(a+1)y}}{a+1} \right] \Big|_0^\infty \\
 &= 1 - \frac{1}{a+1}
 \end{aligned}$$

Differentiation yields that the density function of X/Y is given by

$$f_{X/Y}(a) = 1/(a+1)^2, \quad 0 < a < \infty \quad \blacksquare$$

We can also define joint probability distributions for n random variables in exactly the same manner as we did for $n = 2$. For instance, the joint cumulative probability distribution function $F(a_1, a_2, \dots, a_n)$ of the n random variables $X_1, X_2, \dots, X_n$ is defined by

$$F(a_1, a_2, \dots, a_n) = P\{X_1 \leq a_1, X_2 \leq a_2, \dots, X_n \leq a_n\}$$

If these random variables are discrete, we define their joint probability mass function $p(x_1, x_2, \dots, x_n)$ by

$$p(x_1, x_2, \dots, x_n) = P\{X_1 = x_1, X_2 = x_2, \dots, X_n = x_n\}$$

Further, the n random variables are said to be jointly continuous if there exists a function $f(x_1, x_2, \dots, x_n)$, called the joint probability density function, such that for any set C in n -space

$$P\{(X_1, X_2, \dots, X_n) \in C\} = \int \int_{(x_1, \dots, x_n) \in C} \dots \int f(x_1, \dots, x_n) \, dx_1 \, dx_2 \cdots dx_n$$

In particular, for any n sets of real numbers $A_1, A_2, \dots, A_n$

$$\begin{aligned} P\{X_1 \in A_1, X_2 \in A_2, \dots, X_n \in A_n\} \\ = \int_{A_n} \int_{A_{n-1}} \dots \int_{A_1} f(x_1, \dots, x_n) dx_1 dx_2 \dots dx_n \end{aligned}$$

The concept of independence may, of course, also be defined for more than two random variables. In general, the n random variables $X_1, X_2, \dots, X_n$ are said to be independent if, for all sets of real numbers $A_1, A_2, \dots, A_n$,

$$P\{X_1 \in A_1, X_2 \in A_2, \dots, X_n \in A_n\} = \prod_{i=1}^n P\{X_i \in A_i\}$$

As before, it can be shown that this condition is equivalent to

$$\begin{aligned} P\{X_1 \leq a_1, X_2 \leq a_2, \dots, X_n \leq a_n\} \\ = \prod_{i=1}^n P\{X_i \leq a_i\} \quad \text{for all } a_1, a_2, \dots, a_n \end{aligned}$$

Finally, we say that an infinite collection of random variables is independent if every finite subcollection of them is independent.

EXAMPLE 4.3e Suppose that the successive daily changes of the price of a given stock are assumed to be independent and identically distributed random variables with probability mass function given by

$$P\{\text{daily change is } i\} = \begin{cases} -3 & \text{with probability .05} \\ -2 & \text{with probability .10} \\ -1 & \text{with probability .20} \\ 0 & \text{with probability .30} \\ 1 & \text{with probability .20} \\ 2 & \text{with probability .10} \\ 3 & \text{with probability .05} \end{cases}$$

Then the probability that the stock's price will increase successively by 1, 2, and 0 points in the next three days is

$$P\{X_1 = 1, X_2 = 2, X_3 = 0\} = (.20)(.10)(.30) = .006$$

where we have let X_i denote the change on the i th day. ■